

BG77xA-GL&BG95xA-GL

NIDD Application Note

LPWA Module Series

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About the Document

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-	2021-01-15	Nikola BLAZIC	Creation of the document
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1 Introduction

The support for NIDD (Non-IP Data Delivery) is introduced as of 3GPP Release 13 and later releases. Functions for NIDD may be used to handle MO and MT communication with UE, where the data used for the communication are considered unstructured from the EPS standpoint (which is referred to as Non-IP).

Quectel BG77xA-GL and BG95xA-GL modules support NIDD in NB-IoT RAT, and support both tethered and embedded Non-IP data calls. This document explains how to use the NIDD function of the modules through AT commands.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Model
BG77xA-GL	BG770A-GL
	BG773A-GL
BG95xA-GL	BG950A-GL
	BG951A-GL
	BG953A-GL
	BG955A-GL

2 NIDD AT Commands

2.1. AT Command Introduction

2.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 2: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of the corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

2.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

2.3. Description of NIDD-Related AT Commands

2.3.1. AT+QCFGEXT="nipdcfg" Configure NIDD Connection

This command configures an NIDD connection.

AT+QCFGEXT="nipdcfg" Configure NIDD Connection	
Write Command AT+QCFGEXT="nipdcfg"[,<type>[,<apn>[,<username>,<password>]]]	Response If the optional parameters are omitted, query the current setting. +QCFGEXT: "nipdcfg",<type>,<APN> OK If any of the optional parameters is specified, configure the NIDD connection. OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are not saved.

Parameter

<type>	Integer type. Non-IP outgoing data type. 0 MO Non-IP data type 1 MO Exception Non-IP data type
<apn>	String type. Access point name.
<username>	String type. Selected APN username.

<password> String type. Selected APN password.

NOTE

Before using the selected APN for starting up a Non-IP data call, set the APN PDP type into "Non-IP" with **AT+CGDCONT**. See *document [2]* for detailed information about **AT+CGDCONT**.

2.3.2. AT+QCFGEXT="nipd" Open/Close NIDD Connection

This command opens or closes an NIDD connection.

AT+QCFGEXT="nipd" Open/Close NIDD Connection

Write Command AT+QCFGEXT="nipd",<mode>[,<time out>]	Response If <mode>=0 , close the NIDD connection. OK If <mode>=1 , open an NIDD connection. OK +QIND: "nipd","open",<errcode> If there is an error related to ME functionality: +CME ERROR: <errcode> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are not saved.

Parameter

<mode>	Integer type. Close or open an NIDD connection. 0 Close an NIDD connection. 1 Open an NIDD connection.
<timeout>	Integer type. Timeout for opening the NIDD connection. Parameter is valid only when <mode>=1 . Range: 30–90. Default value: 30. Unit: second.
<errcode>	Integer type. Error code. See <i>Chapter 3</i> for detailed information.

NOTE

1. Non-IP data calls are supported for AP embedded and tethered RmNet calls only.
2. Non-IP data calls are not supported for modem embedded calls.

2.3.3. AT+QCFGEXT="nipds" Send MO Non-IP Data

This command sends MO Non-IP data to a server.

AT+QCFGEXT="nipds" Send MO Non-IP Data

Write Command AT+QCFGEXT="nipds",<mode>,<data>[,<data_length>[,<rai_flag>]]	Response OK If there is an error related to ME functionality: +CME ERROR: <errcode> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are not saved.

Parameter

<mode>	Integer type. Data format. 0 ASCII format string. 1 HEX format string.
<data>	String type. Data to be sent.
<data_length>	Integer type. Length of data to be sent. If this parameter is omitted, <data> can be specified at any length within 1358 bytes of ASCII format. Range: 1–1358. <data> can be specified at any length within 679 bytes of HEX format. Range: 1–679. Unit: byte.
<rai_flag>	Integer type. Whether to enable RAI flag when sending data to network side. 0 Disable 1 Enable
<errcode>	String type. Error code. See Chapter 3 for detailed information.

2.3.4. AT+QCFGEXT="nipdr" Retrieve MT Non-IP Data

This command retrieves the data reported by the URC **+QIND: "nipd","recv"**.

AT+QCFGEXT="nipdr" Retrieve MT Non-IP Data

Write Command AT+QCFGEXT="nipdr"[,<read_length>[,<read_mode>]]	Response +QCFGEXT: "nipdr",<read_actual_length>,<data> OK If there are no data that can be retrieved: +QCFGEXT: "nipdr",0 OK If there is an error related to ME functionality: +CME ERROR: <errcode> If there is any other error: ERROR
Write Command When <read_length> is 0, query the read status of the retrieved data: AT+QCFGEXT="nipdr",0	Response If the connection has existed: +QCFGEXT: "nipdr",<total_receive_length>,<have_read_length>,<unread_length> OK If there is an error related to ME functionality: +CME ERROR: <errcode> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	/

Parameter

<read_length>	Integer type. Length of data to be retrieved. Retrieve all available data if this parameter is omitted. Unit: byte.
<read_mode>	Integer type. Data format. This parameter is valid only when <read_length> is not 0. 0 String type 1 Hex type
<read_actual_length>	Integer type. Actual length of retrieved data. Unit: byte.
<data>	String type. Retrieved data.

<total_receive_length>	Integer type. Total length of received data. Unit: byte.
<have_read_length>	Integer type. Length of retrieved data. Unit: byte.
<unread_length>	Integer type. Length of unread data. Unit: byte.
<errcode>	Integer type. Error code. See Chapter 3 for detailed information.

2.4. Description of NIDD Related URCs

2.4.1. +QIND: "nipd","recv" Indicate Incoming Non-IP Data

After receiving the non-IP data from the MT, the module reports the URC **+QIND: "nipd","recv"** to notify the host about incoming data. Then host can retrieve data via **AT+QCFGEXT="nipdr"**. If the buffer is not empty, and the module receives data again, it will not report a new URC until all the received data have been retrieved via **AT+QCFGEXT="nipdr"** from the buffer. Buffer size: 2048 bytes. If the data received exceed the buffer size, subsequent data will be discarded.

+QIND: "nipd","recv" Indicate the Incoming Data

+QIND: "nipd","recv"	The URC notifies the host about incoming data from the network. Then the host can retrieve the data via AT+QCFGEXT="nipdr" .
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2.4.2. +QIND: "nipd","close" Indicate the Connection is Closed

In case of bad signal or the network goes down, the module reports the URC **+QIND: "nipd","close"** to notify the host that the connection is closed.

+QIND: "nipd","close" Indicate the Connection is Closed

+QIND: "nipd","close"	The URC notifies the host that the connection has been closed accidentally. If the connection closed normally via AT+QCFGEXT="nipd",0 , this URC will not be reported.
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2.5. Example

```

AT+CGDCONT=1,"Non-IP","cmcc"           //Set the PDP type of selected APN into "Non-IP".
OK
AT+CEREG?
+CEREG: 0,1

OK
AT+QCFGEXT="nipdcfg",0,"cmcc"          //Set the Non-IP data type and APN.
OK

```

AT+QCFGEXT="nipdcfg"

+QCFGEXT: "nipdcfg",0,"cmcc"

OK

AT+QCFGEXT="nipd",1,30

//Open an NIDD connection.

OK

+QIND: "nipd","open",0

AT+QCFGEXT="nipds",0,"quectel",7

//Send "quectel" in ASCII format via NIDD connection.

OK

AT+QCFGEXT="nipds",1,"6162636465",10

//Send "abcde" in hex format via NIDD connection.

OK

//When there are any incoming data from the network, the following URC is reported.

+QIND: "nipd","recv"

AT+QCFGEXT="nipdr",0

//Query the retrieved data length.

+QCFGEXT: "nipdr",10,0,10

//There are 10 bytes of data to be retrieved.

OK

AT+QCFGEXT="nipdr",10

//Read 10 bytes of incoming data.

+QCFGEXT: "nipdr",10,0123456789

OK

AT+QCFGEXT="nipdr",0

//Query the retrieved data length.

+QCFGEXT: "nipdr",10,10,0

//All incoming data have been retrieved.

OK

AT+QCFGEXT="nipd",0

//Close the NIDD connection.

OK

3 Summary of <errcode>

The error code **<errcode>** indicates an error related to mobile equipment or network operation. The definition of **<errcode>** is provided in the table below.

Table 3: Summary of <errcode>

<errcode>	Definition
0	Operation successful
651	Invalid input value
652	Send error
653	Network error
654	NIDD busy
655	Timeout error
656	Connection not opened
657	Connection already opened
658	Connection accidentally closed

4 Appendix References

Table 4: Related Documents

Document Name
[1] Quectel_BG77xA-GL&BG95xA-GL_QCFGEXT_AT_Commands_Manual
[2] Quectel_BG77xA-GL&BG95xA-GL_AT_Commands_Manual

Table 5: Terms and Abbreviations

Abbreviation	Description
3GPP	3rd Generation Partnership Project
AP	Application Processor
APN	Access Point Name
ASCII	American Standard Code for Information Interchange
EPS	Evolved Packet System
HEX	Hexadecimal
IP	Internet Protocol
ME	Mobile Equipment
MO	Mobile Originated
MT	Mobile Terminated
NIDD	Non-IP Data Delivery
PDP	Packet Data Protocol
UE	User Equipment
URC	Unsolicited Result Code